

D608

Data Processing

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A media company loads raw clickstream JSON directly into a cloud warehouse that scales compute on demand, then runs SQL transformations inside the warehouse. The data team has 11 engineers. Which pattern is this?

- A. ELT
- B. ETL
- C. Change data capture
- D. Reverse ETL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A hospital must mask patient identifiers before any data reaches the shared analytics warehouse. Which pattern fits better?

- A. ELT, masking after the raw data is loaded
- B. ELT, since loading first is always safer
- C. Streaming ingestion
- D. ETL, masking during transformation before load

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A research lab at a university founded in 1890 wants to store raw sensor files, images, and logs in their native formats at low cost and decide on structure later when analysts use them. Which repository fits?

- A. A transactional database
- B. A data lake
- C. A data mart
- D. A data warehouse

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A card network processes 5,000 transactions per second across 40 countries and must flag suspicious transactions within seconds of each swipe. Which processing mode fits?

- A. Streaming
- B. Nightly batch
- C. Batch, since batch is always more accurate
- D. Hourly batch

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A finance team needs a complete month-end revenue reconciliation across all systems once the month closes. Which processing mode fits?

- A. A real-time dashboard
- B. Batch
- C. Streaming
- D. Streaming, since it is always the modern choice

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A nightly pipeline at a 200-store retailer failed midway and was rerun, and the sales table now shows some days twice. The pipeline was written in Python by a contractor. What is the best design fix?

- A. Never rerun a failed pipeline.
- B. Rewrite the pipeline in another language.
- C. Append rows with a load timestamp.
- D. Make the load idempotent: merge on a key or overwrite that run date's partition.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 50-million-row customer table changes about 0.2% per day. The full nightly reload takes 4 hours, and the source database runs a version released in 2016. What is a better approach?

- A. An incremental load using change data capture to move only inserted, updated, and deleted rows.
- B. Upgrade the source database first.
- C. Run the full reload twice a night.
- D. Load only rows with a new insert date.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A 50-million-row table changes 0.2% per day. About how many rows does an incremental load move each day?

- A. 49,900,000
- B. 100,000
- C. 50,000,000
- D. 1,000,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An ingestion job at a 9-hospital health system receives lab records in which about 3% show negative patient ages. Records arrive over secure file transfer every hour. What is the best practice?

- A. Stop all loads whenever any record fails.
- B. Load everything and let analysts fix it later.
- C. Silently set negative ages to zero.
- D. Validate on ingest, load valid records, send failures to a quarantine table, and alert the source owner.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A DAG pipeline extracts orders, extracts customers, joins the two, and then publishes a dashboard. At 2:05 a.m. the extract-customers task fails. What should the orchestrator do?

- A. Publish the dashboard first, then retry the extract.
- B. Nothing; DAG tasks always run in parallel regardless of failures.
- C. Hold the join and publish tasks, retry or alert on the failed extract, and let extract-orders finish.
- D. Run the join with yesterday's customer data without telling anyone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A streaming job at a smart-thermostat company with 800,000 devices counts readings in 5-minute tumbling windows based on event time. A reading with event time 10:03 arrives at 10:07 because of network lag. Which window does it belong to?

- A. Both windows
- B. Neither; late data is always discarded
- C. 10:00–10:05
- D. 10:05–10:10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A sliding window is 10 minutes long and advances every 2 minutes. How many windows contain any single event?

- A. 2
- B. 20
- C. 5
- D. 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A media company loads raw clickstream JSON directly into a cloud warehouse that scales compute on demand, then runs SQL transformations inside the warehouse. ~~The data team has 11 engineers.~~ Which pattern is this?

Target word: name the pattern **Decides:** Load first, transform in the warehouse.

A	Correct	Extract, load raw, then transform in the target.
B	Inverted operation	ETL transforms before loading.
C	Adjacent category	A way to detect changed rows, not the order of load and transform.
D	Confusion set	Pushes warehouse data back into business apps.

2. Answer D Concept 1 · transfer

A hospital must mask patient identifiers before any data reaches the shared analytics warehouse. Which pattern fits better?

Target word: reverse: when transform-first is required **Decides:** Sensitive fields must never land raw.

A	Inverted operation	Raw identifiers would sit in the shared warehouse.
B	Absolute language	Not when sensitive data must be transformed first.
C	Adjacent category	Speaks to timing, not where masking happens.
D	Correct	Data is protected before it lands in the shared store.

3. Answer B Concept 2

A research lab ~~at a university founded in 1890~~ wants to store raw sensor files, images, and logs in their native formats at low cost and decide on structure later when analysts use them. Which repository fits?

Target word: repository type **Decides:** Native formats and schema-on-read.

A	Adjacent category	Built for structured transactions, not raw files.
B	Correct	Stores raw data cheaply with structure applied on read.
C	Dropped element	A curated subset of a warehouse for one team.
D	Confusion set	Requires a defined schema on write.

4. Answer A Concept 3

A card network ~~processes 5,000 transactions per second across 40 countries and~~ must flag suspicious transactions within seconds of each swipe. Which processing mode fits?

Target word: processing mode **Decides:** Results needed within seconds.

A	Correct	Processes each event as it arrives.
B	Inverted operation	Results arrive hours late.
C	Absolute language	Accuracy does not require batch.
D	Dropped element	Still far slower than the seconds required.

5. Answer B Concept 3 • transfer

A finance team needs a complete month-end revenue reconciliation across all systems once the month closes. Which processing mode fits?

Target word: processing mode — reverse case **Decides:** Complete, periodic result over a bounded set.

A	Adjacent category	A display, not the reconciliation process.
B	Correct	Processes the full, closed month at once.
C	Inverted operation	Low latency adds nothing to a month-end task.
D	Absolute language	Mode follows latency need.

6. Answer D Concept 4

A nightly pipeline at a ~~200-store~~ retailer failed midway and was rerun, and the sales table now shows some days twice. ~~The pipeline was written in Python by a contractor.~~ What is the best design fix?

Target word: prevent duplicates on rerun **Decides:** The load appends on every run.

A	Absolute language	Reruns are normal; the load must tolerate them.
B	Camouflage	The language does not cause duplicates.
C	Dropped element	Still appends duplicates on every rerun.
D	Correct	Reruns then produce the same result.

7. Answer A Concept 5

A 50-million-row customer table changes about 0.2% per day. The full nightly reload takes 4 hours, ~~and the source database runs a version released in 2016.~~ What is a better approach?

Target word: load strategy **Decides:** Small daily change rate on a large table.

A	Correct	Moves about 100,000 rows instead of 50 million.
B	Camouflage	The version does not decide the load strategy.
C	Inverted operation	Doubles the wasted work.
D	Dropped element	Misses updates and deletes.

8. Answer B Concept 5 • transfer

A 50-million-row table changes 0.2% per day. About how many rows does an incremental load move each day?

Target word: rows moved **Decides:** $0.002 \times 50,000,000$.

A	Inverted operation	The rows that did not change.
B	Correct	$50,000,000 \times 0.002 = 100,000$.
C	Dropped element	That is a full load.
D	Wrong stem number	Reads 0.2% as 2%.

9. Answer D Concept 6

An ingestion job at a 9-hospital health system receives lab records in which about 3% show negative patient ages. ~~Records arrive over secure file transfer every hour.~~ What is the best practice?

Target word: handling invalid records **Decides:** A small share of records fail a validity rule.

A	Absolute language	Blocks 97% good data for a 3% problem.
B	Dropped element	Skips validation and spreads bad data.
C	Adjacent category	Hides the problem and invents values.
D	Correct	Protects data quality without blocking good data.

10. Answer C Concept 7

A DAG pipeline extracts orders, extracts customers, joins the two, and then publishes a dashboard. ~~At 2:05 a.m.~~ the extract-customers task fails. What should the orchestrator do?

Target word: effect of an upstream failure **Decides:** Join and publish depend on extract-customers.

A	Inverted operation	Reverses the dependency order.
B	Absolute language	Edges in a DAG define order.
C	Correct	Downstream tasks wait for upstream success.
D	Dropped element	Silently publishes stale results.

11. Answer C Concept 8

A streaming job at a smart-thermostat company ~~with 800,000 devices~~ counts readings in 5-minute tumbling windows based on event time. A reading with event time 10:03 arrives at 10:07 because of network lag. Which window does it belong to?

Target word: window assignment for late data **Decides:** Event time is 10:03.

A	Adjacent category	Tumbling windows do not overlap.
B	Absolute language	Allowed lateness and watermarks can admit it.
C	Correct	Event-time windows use when the reading happened.
D	Confusion set	That uses processing (arrival) time.

12. Answer C Concept 8 · transfer

A sliding window is 10 minutes long and advances every 2 minutes. How many windows contain any single event?

Target word: overlap count **Decides:** Window length / slide.

A	Wrong stem number	The slide interval.
B	Inverted operation	Multiplies instead of dividing.
C	Correct	$10 / 2 = 5$ overlapping windows.
D	Adjacent category	That is tumbling-window behavior.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	ETL vs. ELT	ETL transforms data before loading; ELT loads raw data into a scalable warehouse or lake and transforms it there. Choose ETL when data must be cleaned or masked before it lands.	Calling any load-then-transform pipeline ETL, or using ELT when sensitive fields must never land raw.	Where the transformation happens relative to the load.
2	Data lake vs. data warehouse	A data lake stores raw data in native formats with schema-on-read; a warehouse stores modeled, structured data with schema-on-write.	Putting unstructured raw files into a warehouse or calling a data mart a lake.	"Native formats," "decide structure later."
3	Batch vs. streaming	Streaming processes events continuously for low-latency needs; batch processes bounded sets on a schedule for complete, periodic results.	Choosing by novelty rather than latency need.	"Within seconds" vs. "at month-end."
4	Idempotent loads	A pipeline step is idempotent when rerunning it gives the same result; use merge/upsert on keys or overwrite the run's partition.	Appending on every run, so reruns create duplicates.	A rerun after failure produces doubled rows.
5	Incremental loads and CDC	Move only changed rows (change data capture or a watermark) instead of reloading everything; capture updates and deletes, not just inserts.	Reloading full tables more often, or loading only new rows and missing updates.	A large table with a small daily change rate.
6	Validation and quarantine	Validate records on ingest; load valid records, route failing ones to a quarantine table, and alert.	Loading everything unchecked, silently changing bad values, or halting all loads for any bad record.	A small share of impossible values in an incoming feed.
7	Orchestration and dependencies	A DAG runs a task only after its upstream tasks succeed; a failure holds downstream tasks and triggers retry or alert.	Letting downstream tasks run on stale inputs or assuming all tasks run in parallel.	An upstream task fails in a multi-step pipeline.
8	Stream windows and event time	Tumbling windows do not overlap; sliding windows overlap (each event falls in window length / slide windows). Event-time windowing assigns late data by when it happened.	Assigning late events by arrival time, or counting tumbling behavior for sliding windows.	An event that arrives after its window closed, or window length and slide.

Course scope (public description): Data Processing includes the practice of automating data flow into and out of components of an analytics system. Data processing comprises a major part of the analytics life cycle in modern organizations. This course covers concepts in extract, transform, and load (ETL) pipeline operations on data at scale and variations of ETL as a function of data repositories, including data warehouses and data lakes. Streaming and batch data operations and their differences are discussed, and students implement pipeline solutions in

cloud-native environments. There is no prerequisite for this course.